

Sliding Window Stride vs Overfitting

Stride	Training Windows	Overlap Ratio	Test Pinball	Delta vs Stride=6	Diagnosis
6 [*]	3,523	17%	0.2069	— (optimal)	Sweet spot: enough samples, low overlap
4	3,523	25%	0.2797	-35.2%	Increased overlap, no new information
2	7,101	50%	0.2969	-43.5% (overfit)	99% neighbor overlap, severe overfitting
1	14,089	99%	0.3216	-55.4% (overfit)	Near-identical windows, extreme overfit

GEFCom2014 Zone 1. Stride=1 generates 14K windows but with 99% overlap -> catastrophic generalization failure. Stride=6 is optimal.